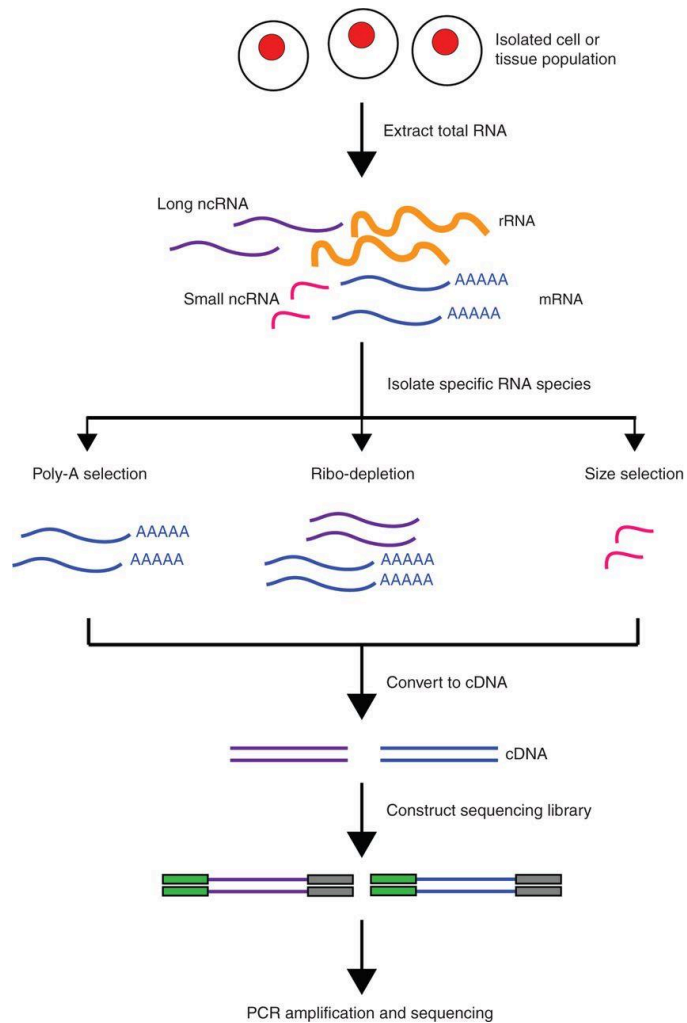


RNA-seq

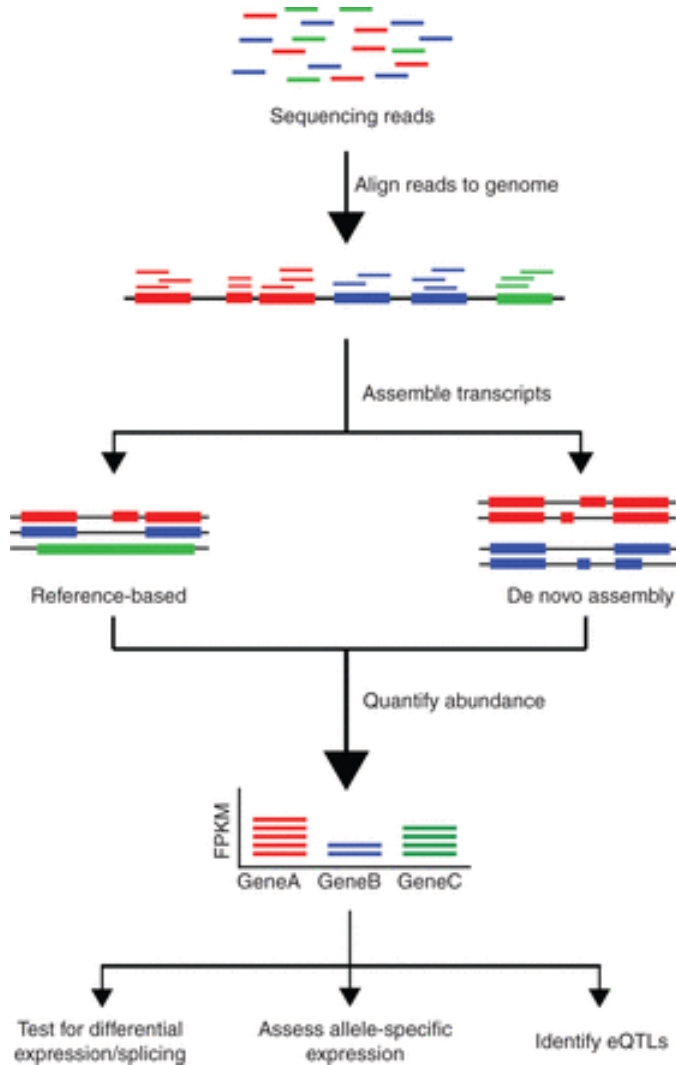
What is RNA-seq and how does it work?

RNA-seq workflow



- RNA is extracted from cell
- Only mRNA (sometimes lincRNA) enriched
- Converted to cDNA
- Construct library
- Sent to sequencing

RNA-seq from sequencing reads to expression



unaligned reads



alignment to genome
(while aware of exon structure)



If we care about alternative transcripts,
we can assemble transcriptome



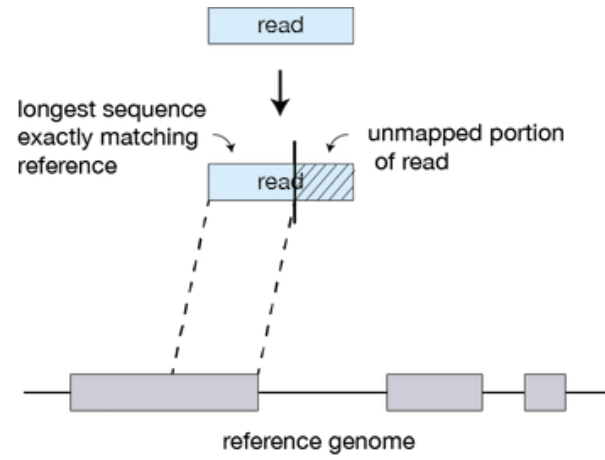
count reads per gene



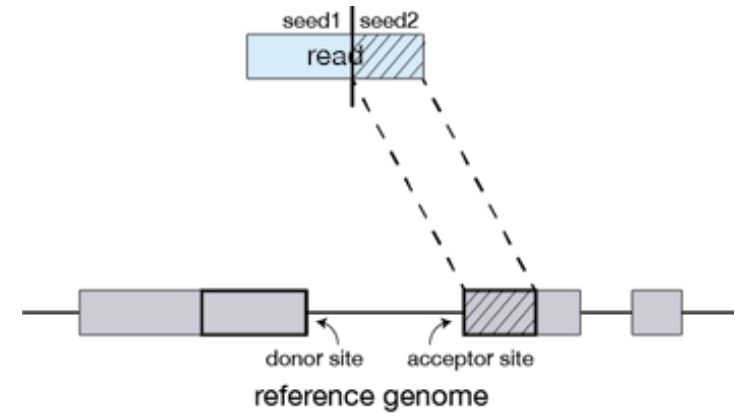
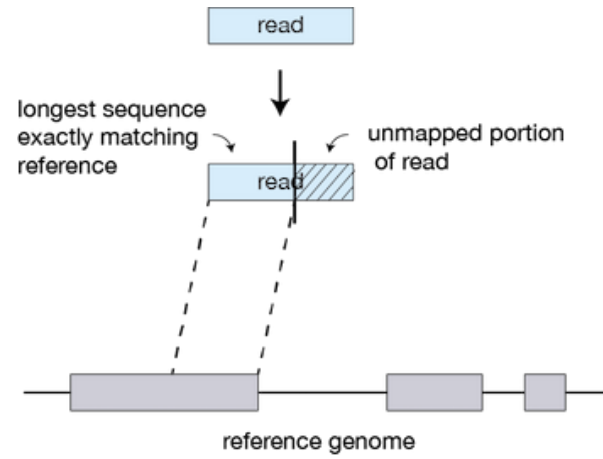
Actual analysis

differential gene expression / allele specific expression / expression QTL

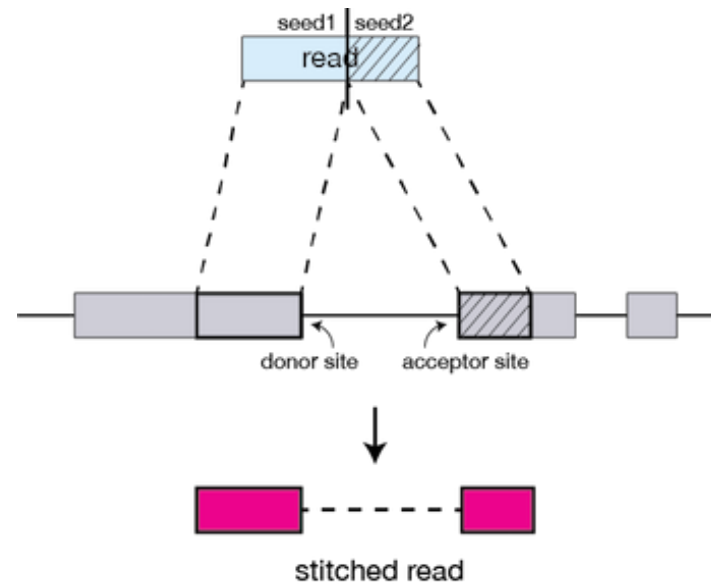
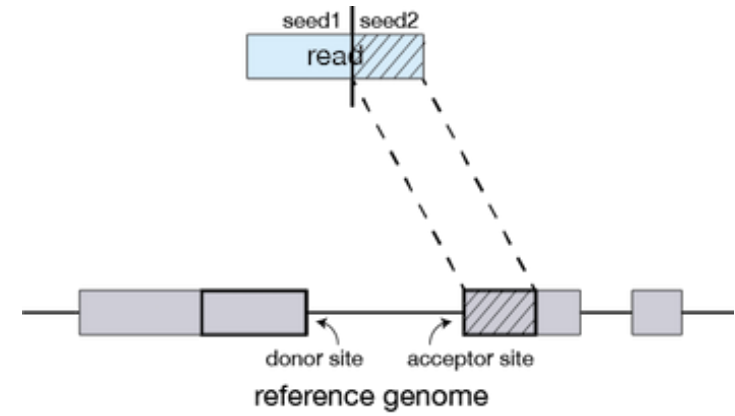
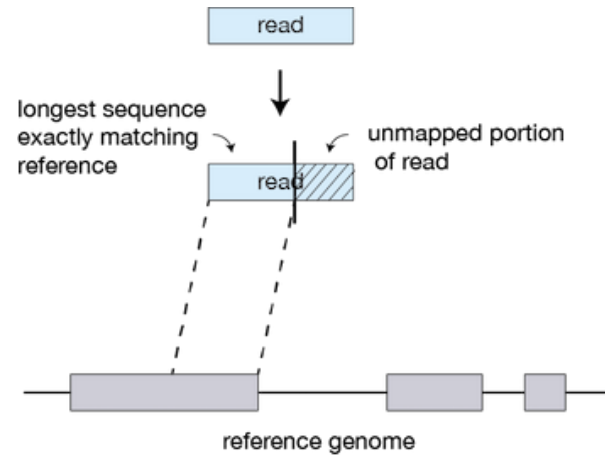
RNA-seq split reads mapping



RNA-seq split reads mapping



RNA-seq split reads mapping



FASTQ format

and sequence reads to map

DNA sequence

```
@EAS139:136:FC706VJ:2:2104:15343:197393 1:Y:18:ATCACG
GTCGTAGTGTCTAGTTCGTAGTCTATGTCGTATTCGTAGGCGTACGTAGTCGTGTAGTCAGTCGTGTT
+
!#!#!!*!#^!#(#*&^*$^#%&$*$&(^&$^^^$)$*&$**$))$*&^$*##$))$)))))
```

EAS139	the unique instrument name
136	the run id
FC706VJ	the flowcell id
2	flowcell lane
2104	tile number within the flowcell lane
15343	x'-coordinate of the cluster within the tile
197393	y'-coordinate of the cluster within the tile
1	the member of a pair, 1 or 2 (<i>paired-end or mate-pair reads only</i>)
Y	Y if the read is filtered, N otherwise
18	0 when none of the control bits are on, otherwise it is an even number
ATCACG	index sequence

Quality scores

$$Q = -10 \log_{10}(P_{\text{err}})$$

RNAseq pipeline

Control1
Control2
Control3
Treatment1
Treatment2
Treatment3

RNAseq pipeline

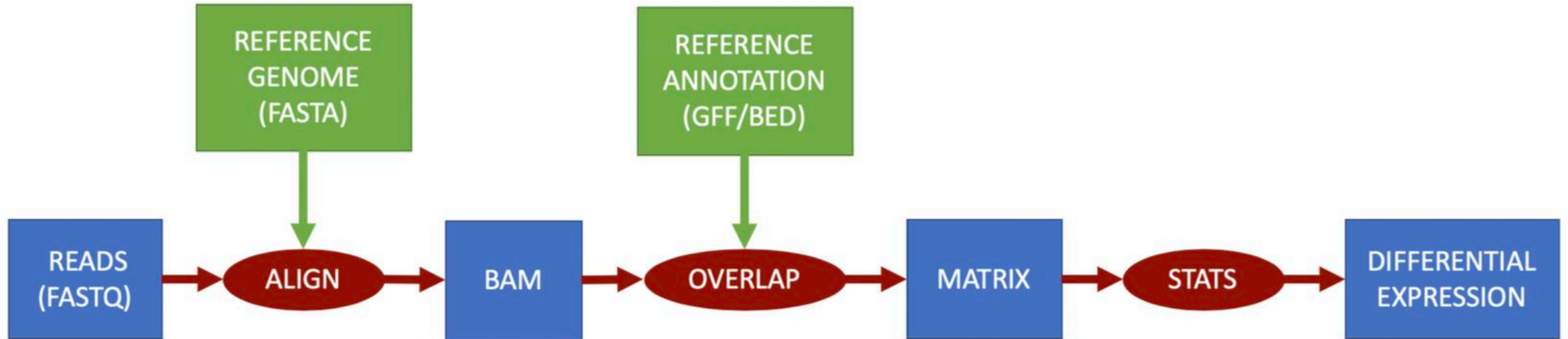
```
#!/usr/bin/env bash
#
# Run fastp on the raw fastq files
#
# -----
set -Eeou pipefail

FQ=/path/to/00_fastq      # Directory containing raw fastq files
SAMPLES=sample-names.txt  # A text file listing basenames of fastq files
OUT=/path/to/put/01_fastp # Where to save the fastp results
THREADS=8

mkdir -p $OUT

for SAMPLE in $(cat $SAMPLES)
do
    fastp -i $FQ/${SAMPLE}_R1.fq.gz \
          -I $FQ/${SAMPLE}_R2.fq.gz \
          -o $OUT/${SAMPLE}.trimmed.1.fq.gz \
          -O $OUT/${SAMPLE}.trimmed.2.fq.gz \
          -h $OUT/${SAMPLE}.fastp.html \
          -j $OUT/${SAMPLE}.fastp.json \
          -w $THREADS
done
```

RNAseq pipeline



RNAseq pipeline

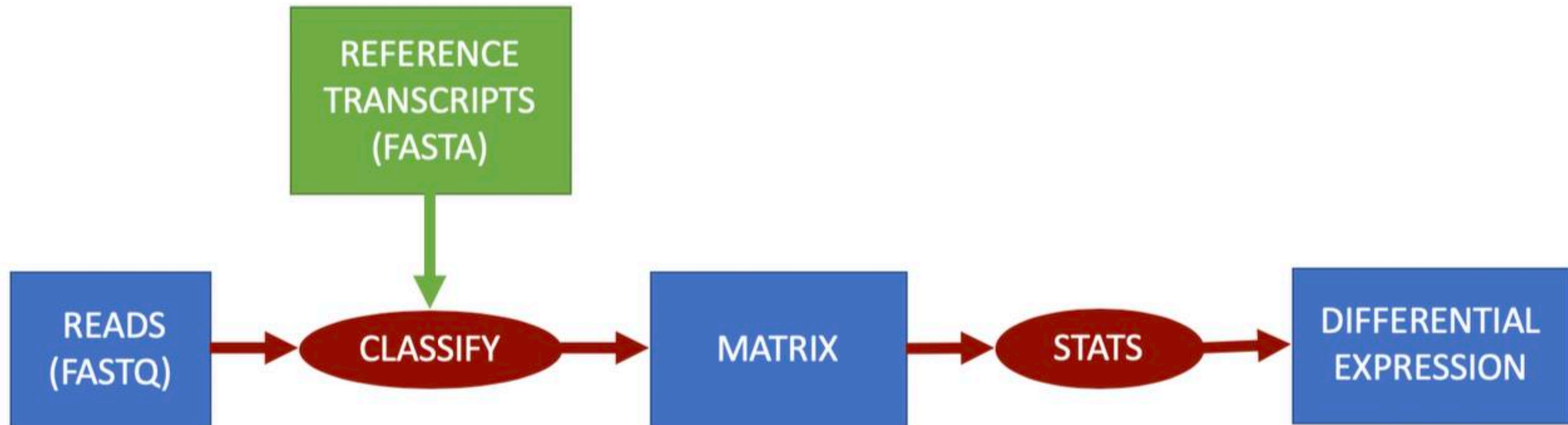
```
#!/usr/bin/env bash
#
# Align reads with STAR
#
# -----
set -Eeou pipefail

SAMPLES=sample-names.txt    # Same sample-names.txt file as above
FQ=/path/to/01_fastp        # Directory containing the fastp output
OUT=/path/to/03_STAR_outs   # Where to save the STAR results
IDX=/path/to/STAR-idx       # Index used by STAR for alignment
THREADS=24

mkdir -p $OUT

for SAMPLE in $(cat $SAMPLES)
do
    STAR --runThreadN $THREADS \
        --genomeDir $IDX \
        --readFilesIn ${FQ}/${SAMPLE}.trimmed.1.fq.gz ${FQ}/${SAMPLE}.trimmed.2.fq.gz \
        --readFilesCommand zcat \
        --outFilterType BySJout \
        --outFileNamePrefix ${OUT}/${SAMPLE}_ \
        --alignSJoverhangMin 8 \
        --alignSJDBoverhangMin 1 \
        --outFilterMismatchNmax 999 \
        --outFilterMismatchNoverReadLmax 0.04 \
        --alignIntronMin 20 \
        --alignIntronMax 1000000 \
        --alignMatesGapMax 1000000 \
        --outMultimapperOrder Random \
        --outSAMtype BAM SortedByCoordinate \
        --quantMode GeneCounts;
done
```

RNAseq pipeline



RNAseq pipeline

```
#!/usr/bin/env bash
#
# Perform transcript quantification with Salmon
#
# -----
set -Eeou pipefail

SAMPLES=sample-names.txt # Same sample-names.txt file as above
IDX=/path/to/salmon-idx   # Index used by Salmon
FQ=/path/to/01_fastp      # Directory containing the fastp output
OUT=/path/to/02_quants    # Where to save the Salmon results
THREADS=12

mkdir -p $OUT

for SAMPLE in $(cat $SAMPLES)
do
    salmon quant \
        -i $IDX \
        -l A \
        -1 $FQ/${SAMPLE}.trimmed.1.fq.gz \
        -2 $FQ/${SAMPLE}.trimmed.2.fq.gz \
        --validateMappings \
        --gcBias \
        --seqBias \
        --threads $THREADS \
        -o $OUT/${SAMPLE}_quants
done
```



(Some) of the Many Gene IDs

- Ensembl ids
 - ENS[species prefix][feature type prefix][a unique eleven digit number].[version number]
 - Human: **ENS**G00000168769, Mouse: **ENSMUST**00000101509.9
- HUGO Gene Naming Consortium sets gene names like BRCA1
- Other ids:
 - Entrez
 - UCSC
 - NCBI
 - RefSeq
 - OMIM
 - **Uniprot**
- If you don't know what type of id it is, google the id

Gene name errors are widespread in the scientific literature

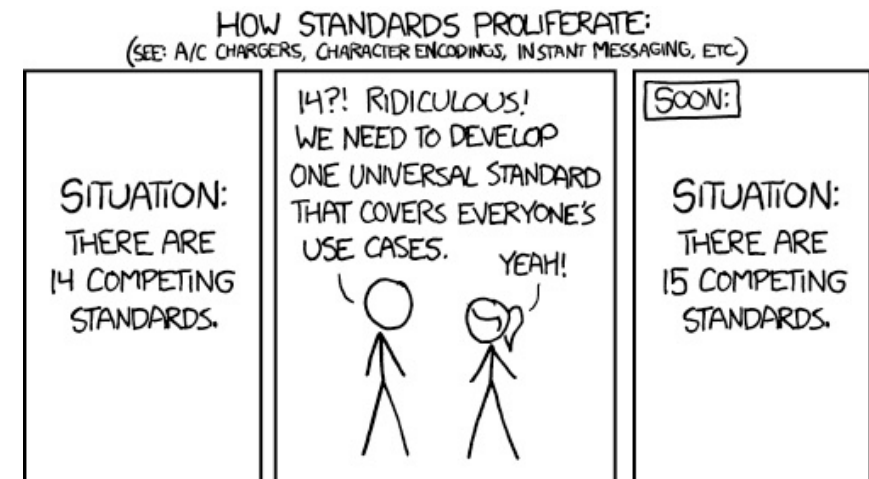
[Mark Ziemann](#), [Yotam Eren](#) & [Assam El-Osta](#) ✉

[Genome Biology](#) 17, Article number: 177 (2016) | [Cite this article](#)

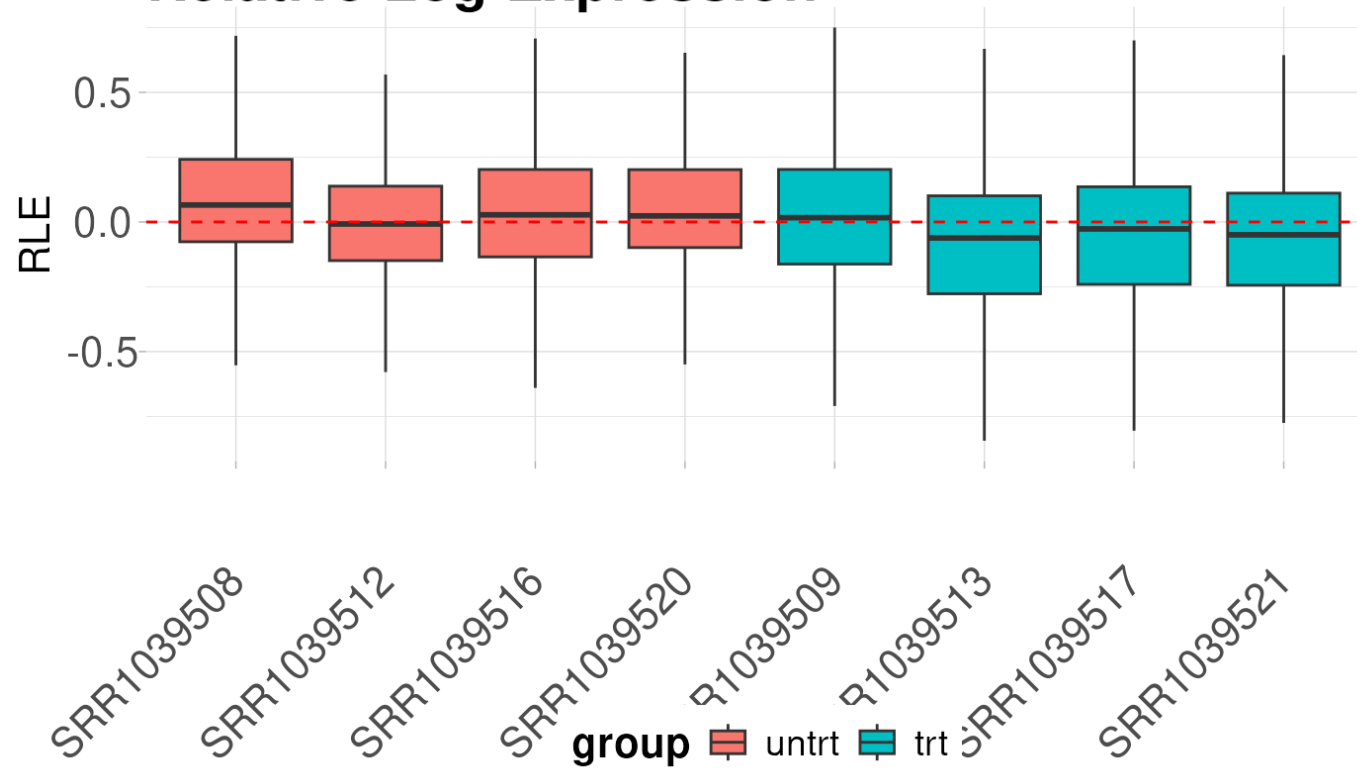
134k Accesses | 53 Citations | 3209 Altmetric | [Metrics](#)

Abstract

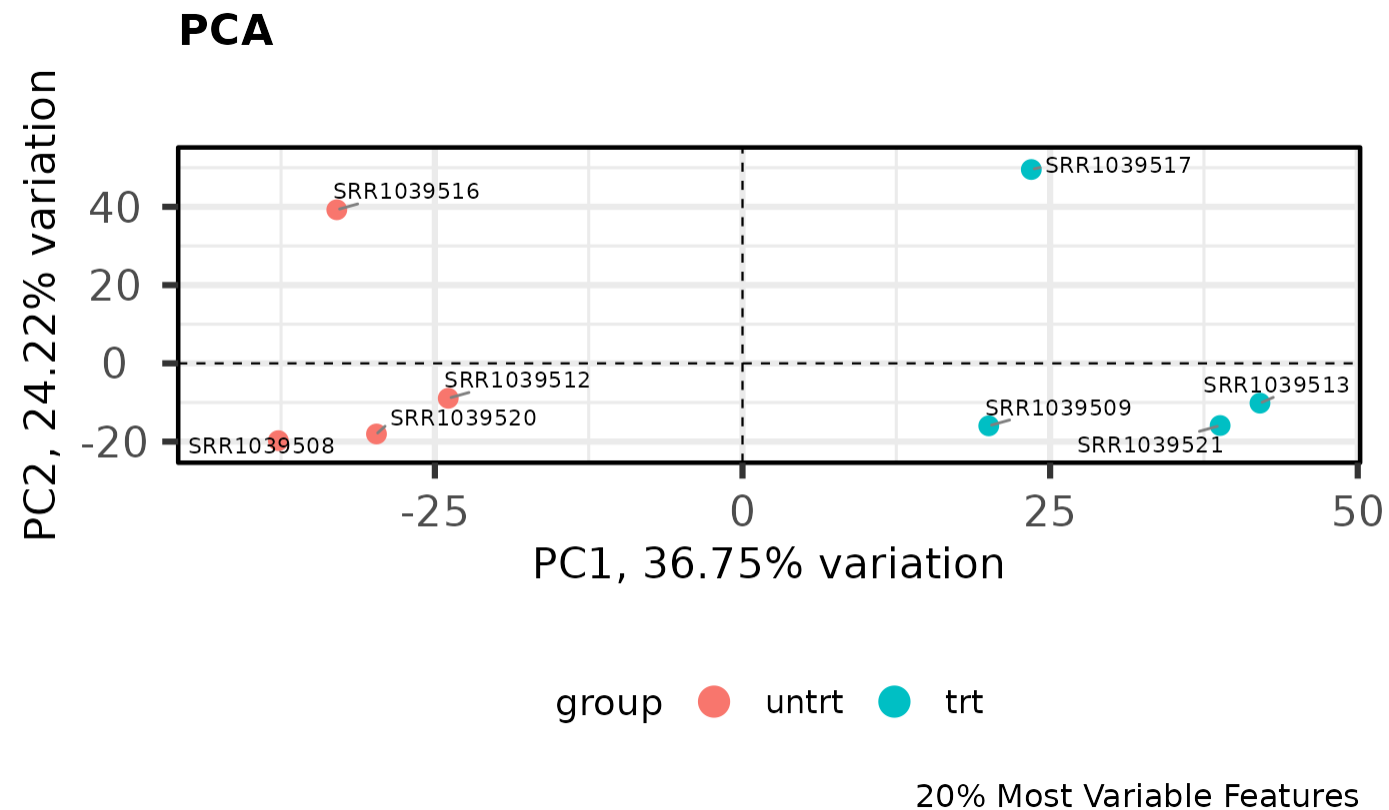
The spreadsheet software Microsoft Excel, when used with default settings, is known to convert gene names to dates and floating-point numbers. A programmatic scan of leading genomics journals reveals that approximately one-fifth of papers with supplementary Excel gene lists contain erroneous gene name conversions.



Relative Log Expression

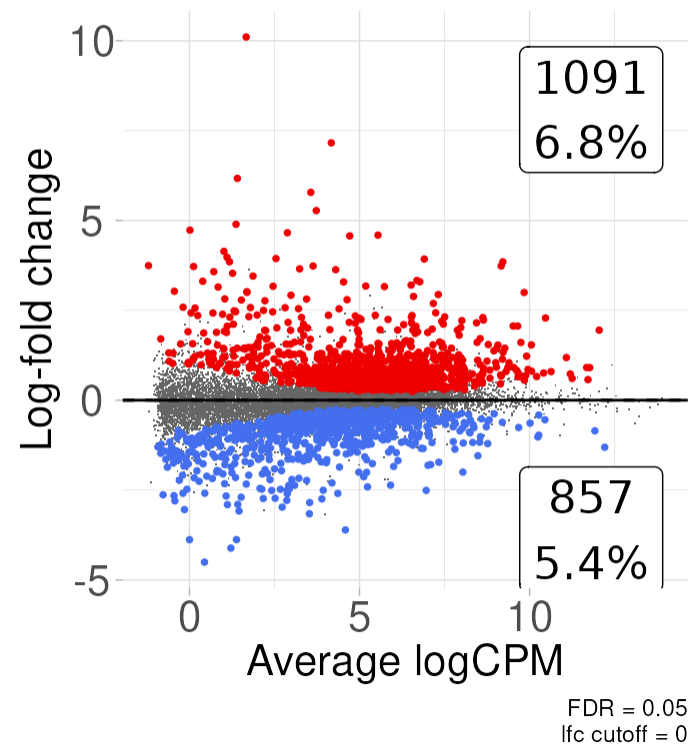
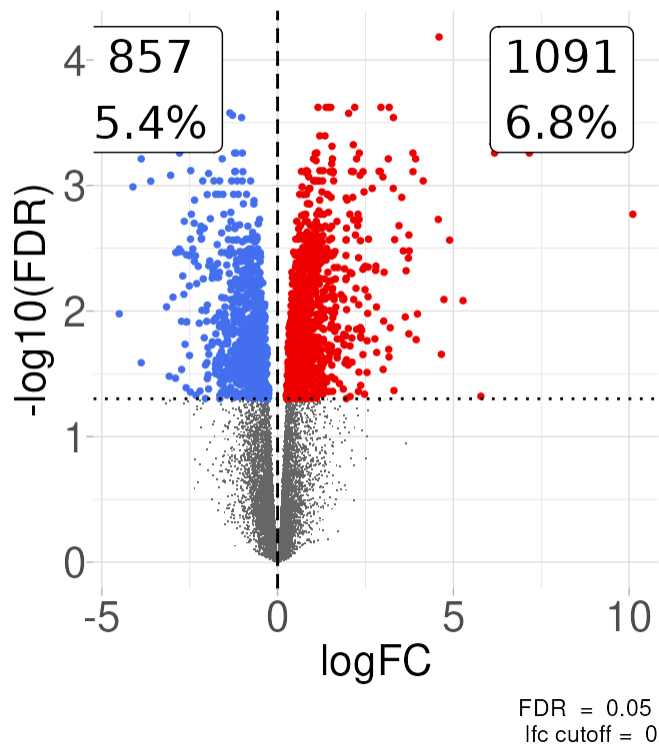


Plotting DE results: PCA



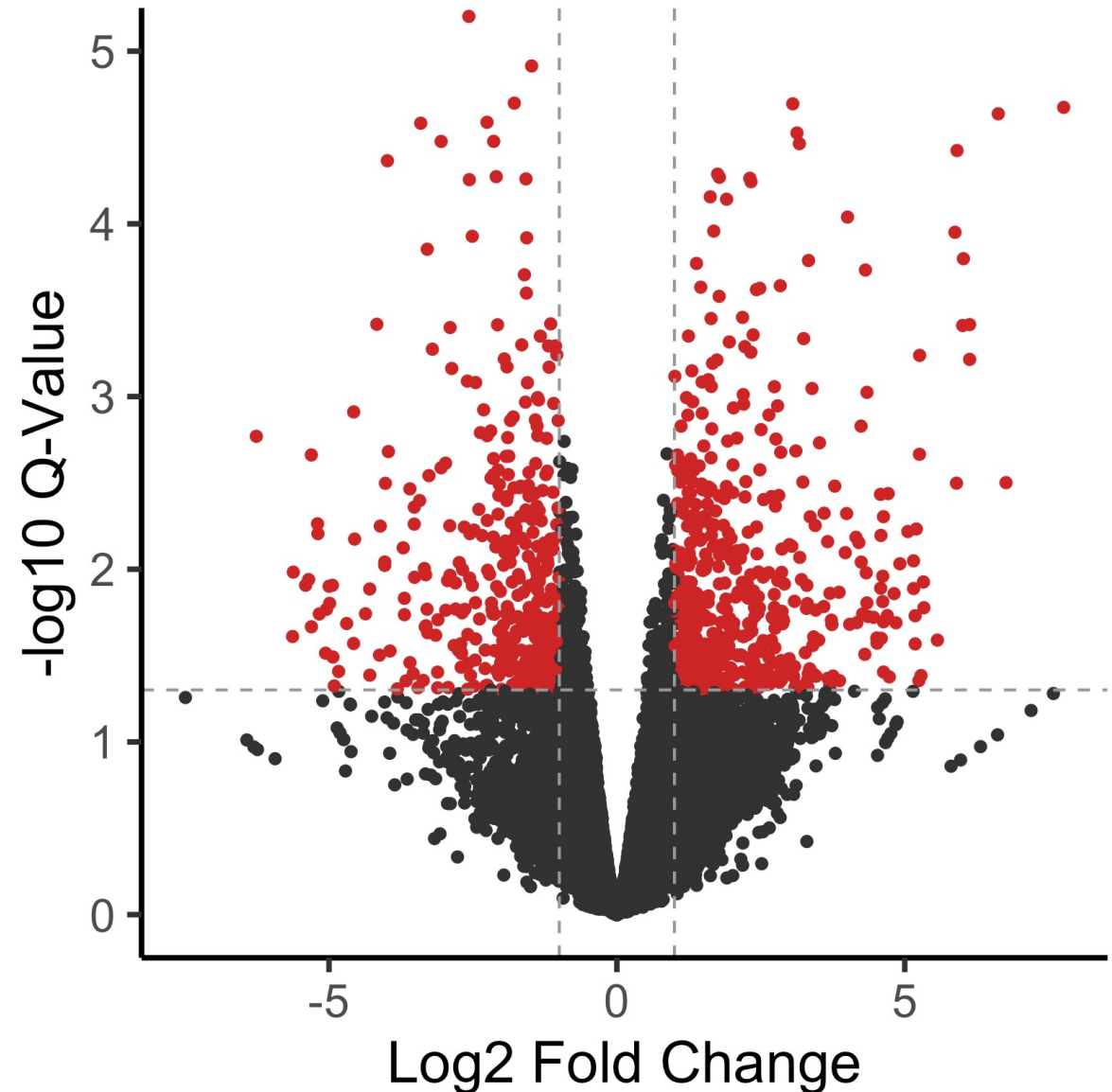
Plotting DE results: Volcano and MA plot

Treatment vs. Control



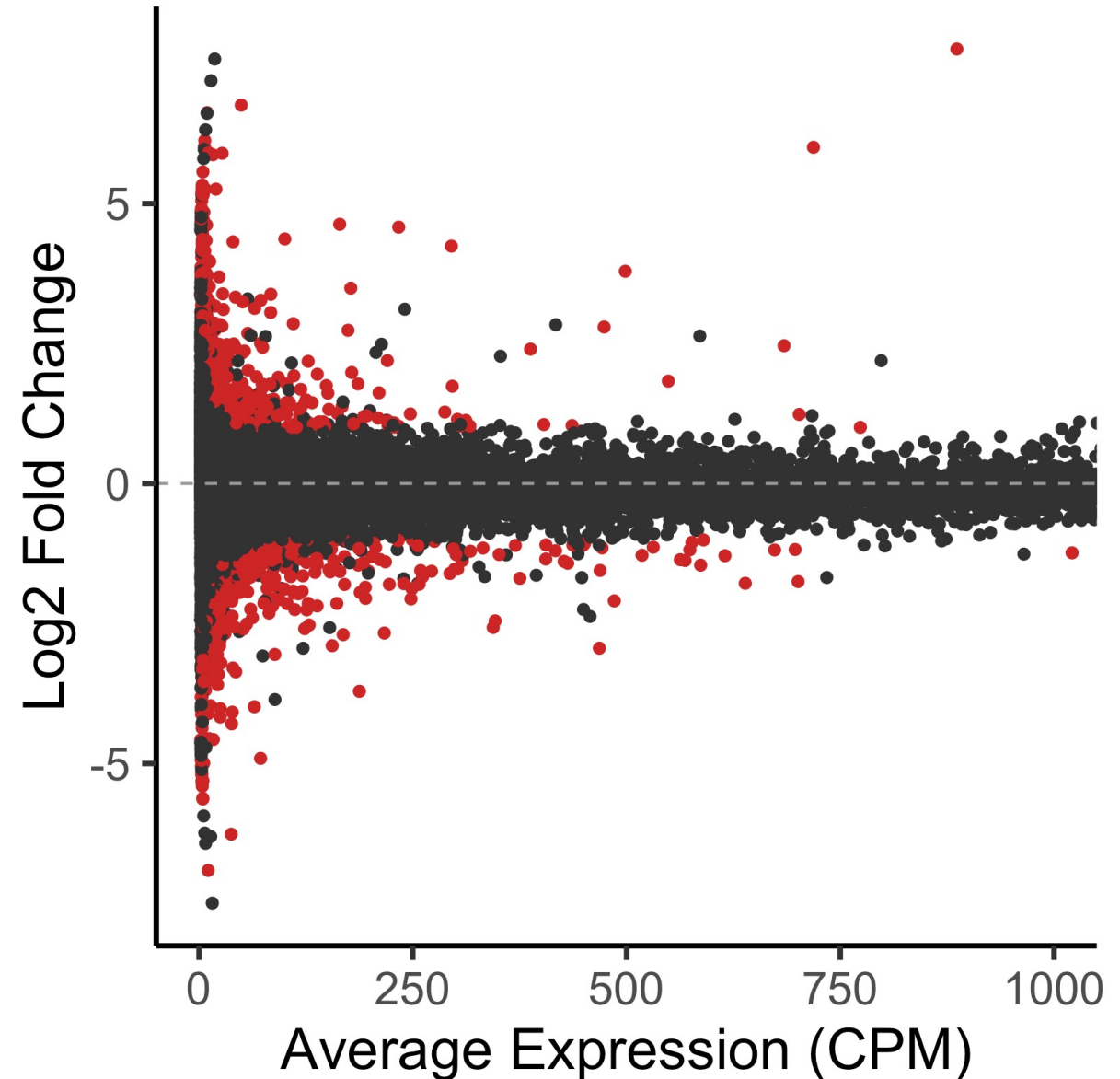
Volcano Plot

- X-axis shows the average log2 fold change for the variable of interest
- Y-axis negative log10 corrected p-value
- Overall shows how the expression in the treatment group changes and how statistically significant the changes are

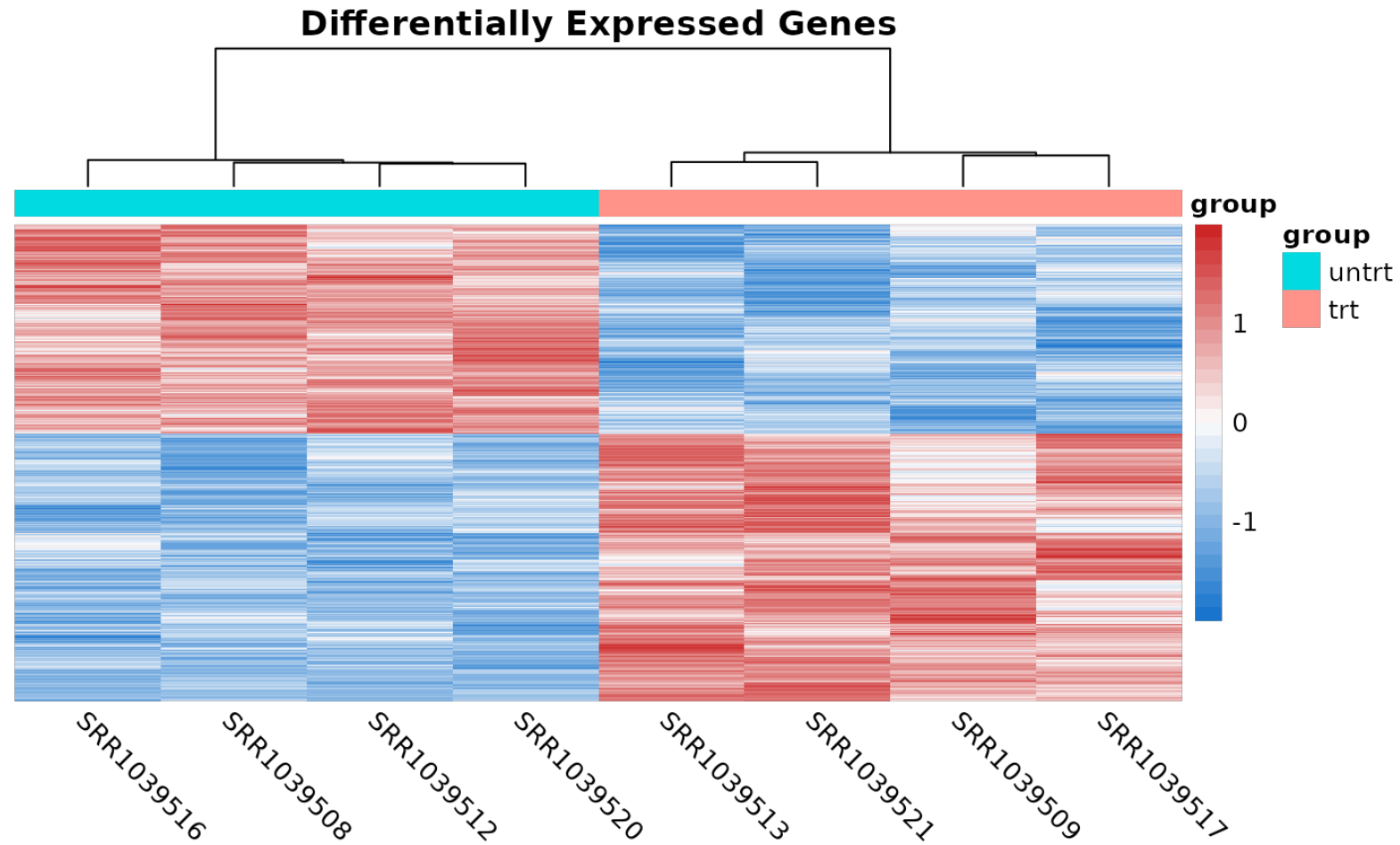


MA Plot

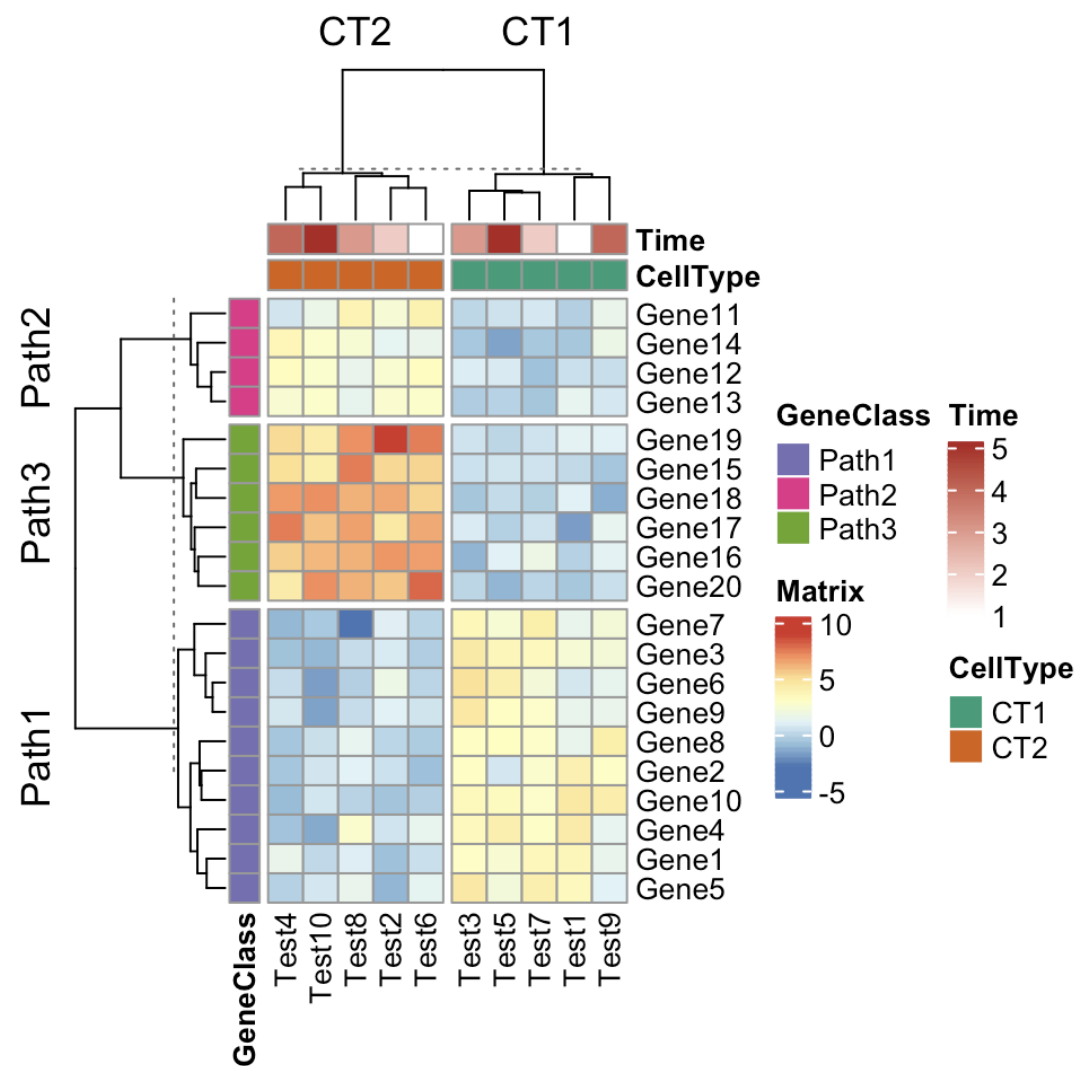
- X-axis shows the average expression for all samples
- Y-axis shows the log2 fold change for the variable of interest
- Overall shows how the expression in the variable group changes relative to all other samples



Plotting DE results: Heatmap



Heatmap



UpSet Plot

- New variation on a venn diagram because venn diagrams are not great at visualizing more than 2 or 2 sets
- https://en.wikipedia.org/wiki/Venn_diagram
- Points and line on the bottom show the overlap between the conditions on the left
- Bar height and counts show the number of features that overlap

B

